

THOMAS FARINELLI

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SUMMARY

My academic work has focused on Machine Learning, Deep Learning, biomedical data analysis, medical image processing, eye-tracking data, statistical analysis, and the development of data-processing pipelines. I am particularly interested in the use of Artificial Intelligence to support medical diagnosis, clinical decision-making, biomarker identification, and the analysis of complex biomedical data.

EDUCATION

Roma Tre University 2024-2026

Master's Degree in Computer Engineering - Artificial Intelligence and Machine Learning

Current weighted average: 28.19/30.

Degree-award average according to the degree programme regulations: 106.42/110 (equivalent to 29.02/30).

Credits earned: 94/120 CFU. Expected graduation: October 2026.

Master's Thesis: Analysis of the Correlation Between Visual Impairments and Eye-Tracking Data.

Supervisor: Prof. Giuseppe Sansonetti. Co-supervisor: Prof. Marco Barbieri.

The thesis investigates presbyopia and eye-tracking data collected during reading tasks using Tobii Pro Glasses 3. A dedicated processing pipeline filters raw gaze data, identifies reading sequences, extracts fixations, saccades and regressions, and supports descriptive and inferential statistical comparison with a normative control group.

Roma Tre University 2019-2024

Bachelor's Degree in Computer Engineering (L-8 - Information Engineering)

Final grade: 103/110. Credits: 180 CFU. Graduation date: 21 March 2024.

Bachelor's Thesis: Machine Learning Techniques for the Diagnosis of Pulmonary Diseases. The work explored data-driven methods for pulmonary-disease diagnosis and clinical decision support, including Naive Bayes, SVM, Decision Trees, XGBoost, KNN and Logistic Regression.

RESEARCH AND ACADEMIC PROJECTS

Fine-Tuning Small Language Models for Structured Information Extraction | *Deep Learning project*

- Fine-tuned Phi-3 (3.5B), Mistral (7B) and Qwen 2.5 (1.5B) on a subset of RecipeNLG to transform unstructured text into rigid JSON representations.
- Used LoRA for Phi-3 and Qwen 2.5, QLoRA for Mistral, and LLaMA 3.3 70B as an inference-only baseline.
- Evaluated syntactic validity, schema compliance, semantic correctness and generalization to unseen examples; fine-tuning produced substantial improvements, particularly for Qwen 2.5.

Systematic Study of Preprocessing Techniques for Histopathological Images Using ResNet-18 | *Medical Imaging and Deep Learning project*

- Team research project on approximately 25,000 lung and colon histopathology images, including benign tissue, adenocarcinoma and squamous cell carcinoma.
- Compared seven preprocessing strategies and both ImageNet-pretrained and from-scratch ResNet-18 configurations.
- Personal contribution: Wavelet preprocessing, Retinex and Sobel filtering, CLAHE, and experimentation with pre-trained neural networks.
- Evaluation included accuracy, precision, recall, F1-score, confusion matrices, t-SNE, training behaviour, false-negative analysis and Grad-CAM; Wavelet preprocessing with pre-trained ResNet-18 reached approximately 99.72% accuracy with zero false negatives.

Oculomotor Movement Analysis | *Master's thesis research code*

- Developed procedures for processing gaze recordings and extracting fixations, saccades, regressions and reading-related oculomotor events.

- Analysed variations associated with font size and visual difficulty and compared normative and presbyopic participants using descriptive and statistical methods.

Portfolio Website | <https://thomas930821.github.io/>

- Academic projects and implementations in Artificial Intelligence, Deep Learning, Machine Learning, biomedical applications and data analysis.

TECHNICAL SKILLS

- Artificial Intelligence and Machine Learning: supervised and unsupervised learning, neural networks, CNNs, transfer learning, model fine-tuning, Small and Large Language Models, LoRA, QLoRA, structured information extraction, model evaluation.
- Biomedical AI and Medical Data Analysis: medical image classification, histopathological image analysis, eye-tracking and oculomotor analysis, biomedical data processing, clinical decision-support applications, biomarker-oriented analysis.
- Data Analysis and Statistics: data preprocessing, descriptive and inferential statistics, non-parametric tests, correlation analysis, dimensionality reduction, t-SNE, experimental data analysis, computational pipelines.
- Explainable AI and Image Processing: Grad-CAM, CLAHE, Wavelet transforms, Retinex, Sobel filtering, image augmentation and preprocessing.
- Programming and Scientific Computing: Python, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Jupyter, Git and GitHub.
- Research practice: scientific literature review, experimental design, critical thinking, quantitative analysis, interpretation of results, scientific and technical writing, reproducible computational workflows, interdisciplinary teamwork.

MASTER'S DEGREE COURSEWORK

Artificial Intelligence - 29/30 (9 CFU); Software Systems Architecture - 28/30 (9 CFU); Automata, Languages and Computing - 19/30 (9 CFU); Machine Learning - 30/30 (9 CFU); Artificial Intelligence from Engineering to Arts - 30/30 (6 CFU); Technologies and Architectures for Data Management - 30/30 with honours (9 CFU); Computer Graphics - 30/30 (6 CFU); Deep Learning - 30/30 with honours (6 CFU); Decision Support Systems and Analytics - 25/30 (6 CFU); Algorithms for Big Data - 28/30 (6 CFU); Intelligent Systems for the Internet - 30/30 with honours (6 CFU); Automated Planning - 30/30 with honours (6 CFU); Data Law - 30/30 (6 CFU); Skills Useful for Entering the Labour Market - Pass (1 CFU); Final Examination / Master's Thesis - 26 CFU, in progress.

BACHELOR'S DEGREE COURSEWORK

Mathematical Analysis I - 18/30 (12 CFU); Geometry and Combinatorics - 30/30 (12 CFU); Fundamentals of Computer Science - 25/30 (12 CFU); Chemistry - 24/30 (6 CFU); Algorithms and Data Structures - 30/30 with honours (9 CFU); Discrete-Event Systems Analysis - 24/30 (6 CFU); Electrical Engineering and Electronics - 28/30 (9 CFU); Computer Architecture - 23/30 (6 CFU); Operations Research I - 24/30 (6 CFU); Fundamentals of Automatic Control - 27/30 (9 CFU); Functional Programming - 26/30 (6 CFU); Database Systems I - 25/30 (6 CFU); Computer Networks - 22/30 (6 CFU); Object-Oriented Programming - 22/30 (9 CFU); Fundamentals of Telecommunications - 22/30 (9 CFU); Mobile Computing - 30/30 with honours (6 CFU); Engineering Economics - 29/30 (6 CFU); Web Information Systems - 22/30 (6 CFU); Artificial Intelligence and Machine Learning - 27/30 (6 CFU); Physics I - 24/30 (12 CFU); Software Analysis and Design - 27/30 (6 CFU); Internship - Pass (9 CFU); Final Examination - Pass (3 CFU). Bachelor's weighted average: 25.11/30.

INTERNATIONAL EXPERIENCE

London, United Kingdom (2017-2019). Living in London provided significant experience in an international and multicultural environment and strengthened independence, adaptability, communication skills and the ability to interact with people from different cultural and professional backgrounds.

REFERENCES

Academic references available upon request. Prof. Giuseppe Sansonetti - Roma Tre University, Department of Civil, Computer Science and Aeronautical Technologies Engineering. Prof. Fabio Gasparetti - Roma Tre University, Department of Civil, Computer Science and Aeronautical Technologies Engineering.